

Mohamed Mostafa

Software Engineer

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SUMMARY

Backend Engineer with 4+ years of building and operating production payment systems and real-time services in Node.js and TypeScript. Shipped multi-provider payment integrations (PayMob, Amazon Payment Services, Stripe), event-driven background processing, and SSE/WebSocket notifications across two live marketplace platforms. Focused on reliability, idempotency, and performance under real transaction volume.

EXPERIENCE

Backend Engineer | **Lesoll** Jun 2023 – Present · Cairo, Egypt · On-site

- Built and operated the payment system integrating PayMob and Amazon Payment Services, processing **~10K transactions/month** with HMAC-verified webhooks and idempotent handlers that guarantee zero double-charges.
- Designed event-driven background processing (BullMQ + Redis) for email/SMS/OTP and notifications, moving **~500 jobs/day** of slow third-party calls off the request path with automatic retries — removing external-provider latency and failures from user-facing endpoints.
- Engineered a real-time chat and notifications layer (WebSockets / SSE) serving **5k concurrent users** with reliable delivery.
- Eliminated N+1 query patterns and refactored MongoDB aggregation pipelines, reducing key endpoint response times from **680 ms to 230 ms**.
- Built the CoHost system, cancellation policy, booking, reports, and employee tooling, plus a Puppeteer traffic-scraping pipeline used by **4k internal users**.
- Owned containerised deployment on AWS — Lesoll via Docker + EC2, EGYStay via Docker + EKS — with GitHub Actions CI/CD and Nginx, reducing deploy time to **1 minute**.

PROJECTS

Gosha — AI Prompt Optimisation Engine 

- Architected a self-learning prompt optimisation engine that rewrites raw user input into structured, model-specific prompts (Claude XML / GPT Markdown) using a weighted rule system — zero external AI API calls.
- Designed a 5-phase pipeline (Analyse → Learn → Transform → Merge → Record) with a feedback loop that adjusts rule weights from user ratings, enabling category-specific personalisation over time.
- Built a full SaaS backend with PASETO/JWT auth, tiered subscription billing (Free → Enterprise), dual payment integration (Stripe international + PayMob MENA card & wallet), and a daily token ledger per plan.
- Implemented real-time notifications via SSE backed by Redis pub/sub, async email jobs with BullMQ, structured logging with Pino, and a Prometheus metrics endpoint for observability.

gen-import — TypeScript Barrel File Generator

- Built a zero-dependency CLI that auto-generates barrel files (index.ts) for Node.js/TypeScript projects using the TypeScript compiler API for accurate AST-level export detection.
- Implemented an --init command that injects tsconfig path aliases and package.json scripts idempotently, with --alias support for custom path prefixes.
- Added --watch mode using native fs.watch (no chokidar), enabling real-time regeneration with zero runtime dependencies; separates type-only from value exports and supports CommonJS and ESM.

EDUCATION

Bachelor of Computer Science · Modern Academy – Maadi, Cairo, Jul 2022

TECHNICAL SKILLS

Backend: Node.js, Express.js, TypeScript, RESTful APIs, WebSockets, SSE

Databases & Caching: MongoDB, Mongoose, PostgreSQL, Prisma, Redis

Messaging & Async: BullMQ, message queues, Redis pub/sub

Payments: PayMob, Amazon Payment Services, Stripe (webhooks, HMAC, idempotency)

Cloud & DevOps: AWS (EC2, EKS, S3, ECR), Docker & Docker Compose, GitHub Actions CI/CD

Frontend: Vue.js, Vite, TailwindCSS, UnoCSS